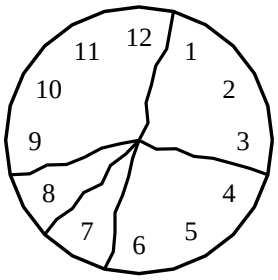
Preview: set-03/by_level/Grade_03-04/y2002_q11 (v1)

Sums of numbers around a circle: use the total, split into equal/consecutive parts

Part A: Each question is worth 3 points

1. The face of a clock is cracked into 5 pieces as shown.

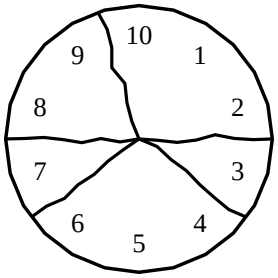


Which piece has the smallest sum of numbers?

A. the piece with the 7 ***B.*** the piece with the 8 ***C.*** the piece with the 12

D. the piece with the 3 ***E.*** the piece with the 6

2. A round cake with the numbers 1 to 10 around its edge is cracked into 5 pieces as shown.



Which piece has the smallest sum of numbers?

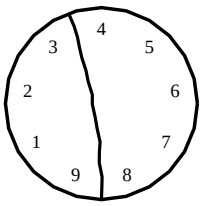
A. the piece with the 3 ***B.*** the piece with the 6 ***C.*** the piece with the 7

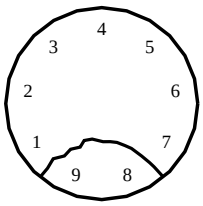
D. the piece with the 9 ***E.*** the piece with the 10

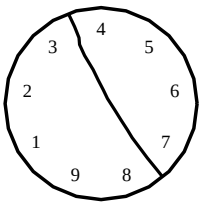
Part B: Each question is worth 4 points

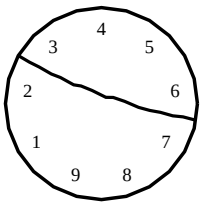
3. A round cookie with the numbers 1 to 9 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.

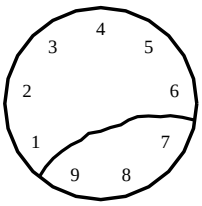
Which picture shows how it is cracked?

A. 

B. 

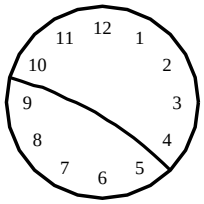
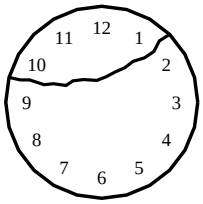
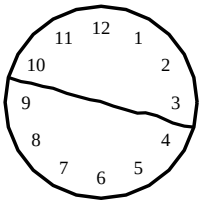
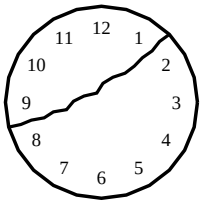
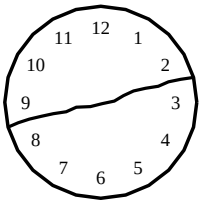
C. 

D. 

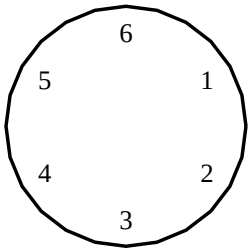
E. 

4. The face of a clock is cracked into 2 pieces. All the pieces have the same sum of numbers.

Which picture shows how it is cracked?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

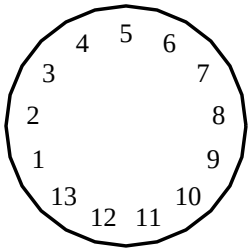
5. A round cookie with the numbers 1 to 6 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.



What is the largest of these sums?

A. 7 ***B.*** 9 ***C.*** 10 ***D.*** 11 ***E.*** 12

6. A round cookie with the numbers 1 to 13 around its edge is cracked into 2 pieces. The sums of the numbers in the pieces are two consecutive numbers.

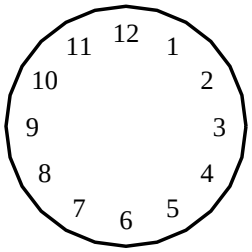


What is the smallest of these sums?

A. 49 ***B.*** 47 ***C.*** 46 ***D.*** 45 ***E.*** 44

Part C: Each question is worth 5 points

7. The face of a clock is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 11 and 12, and between 5 and 6

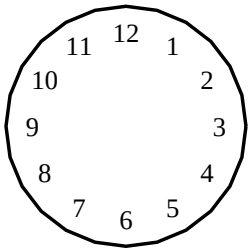
B. between 5 and 6, and between 10 and 11

C. between 3 and 4, and between 9 and 10

D. between 1 and 2, and between 7 and 8

E. between 2 and 3, and between 7 and 8

8. The face of a clock is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 4 and 5, and between 7 and 8

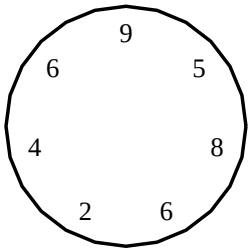
B. between 11 and 12, and between 1 and 2

C. between 3 and 4, and between 9 and 10

D. between 3 and 4, and between 10 and 11

E. between 1 and 2, and between 9 and 10

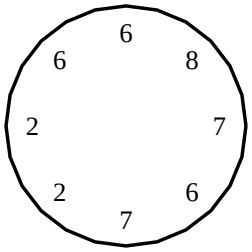
9. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. Numbers are written around a round plate as shown. Tom wants to break the plate with one straight line so that the two pieces have the same sum of numbers.



In how many different ways can he do it?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

Answer Key

1. **D**	**2. A**	**3. C**	**4. C**	**5. D**	**6. D**
7. **C**	**8. C**	**9. A**	**10. B**		

Solutions

1. D (the piece with the 3) — The sums are: 1+2+3 = 6, 4+5+6 = 15, 7 = 7, 8 = 8, 9+10+11+12 = 42. [set-03/by_level/Grade_03-04/2002 Q11 · biggest_piece]

2. A (the piece with the 3) — The sums are: 10+1+2 = 13, 3 = 3, 4+5+6 = 15, 7 = 7, 8+9 = 17. [set-03/by_level/Grade_03-04/2002 Q11 · biggest_piece]

3. C — All numbers add up to 45, so the pieces must be 22 + 23 = 45. Only this picture gives 22, 23. [set-03/by_level/Grade_03-04/2002 Q11 · which_picture]

4. C — All numbers add up to 78, so the pieces must be 39 + 39 = 78. Only this picture gives 39, 39. [set-03/by_level/Grade_03-04/2002 Q11 · which_picture]

5. D (11) — All the numbers add up to 21. Two consecutive numbers that add up to 21: 10 + 11 = 21. The largest is 11. [set-03/by_level/Grade_03-04/2002 Q11 · largest_sum]

6. D (45) — All the numbers add up to 91. Two consecutive numbers that add up to 91: 45 + 46 = 91. The smallest is 45. [set-03/by_level/Grade_03-04/2002 Q11 · largest_sum]

7. C (between 3 and 4, and between 9 and 10) — Total 78, so each piece has 39: 4+5+6+7+8+9 = 39 and 10+11+12+1+2+3 = 39. [set-03/by_level/Grade_03-04/2002 Q11 · straight_crack]

8. C (between 3 and 4, and between 9 and 10) — Total 78, so each piece has 39: 4+5+6+7+8+9 = 39 and 10+11+12+1+2+3 = 39. [set-03/by_level/Grade_03-04/2002 Q11 · straight_crack]

9. A (1) — Each half must have 20. The ways: 8+6+2+4 | 6+9+5. [set-03/by_level/Grade_03-04/2002 Q11 · count_ways]

10. B (1) — Each half must have 22. The ways: 7+6+7+2 | 2+6+6+8. [set-03/by_level/Grade_03-04/2002 Q11 · count_ways]